Department of Mathematics
National Institute of Technology Kurukshetra
Statistical Distribution (MAIC 205)

Time: 06:00 pm - 06:50 pm | Max Marks: 15 | Date: 25/09/25

1. (a) Suppose that two dice are rolled, and let the random variable X denotes the sum of the points.
   1. Find the probability mass function (PMF) for X.
   2. Find the graph of its CDF.

   (2+2+2)

   (b) If the weather is good (which happens with probability 0.6), a student walks the 2 km to class at a speed of $V = 5$ km per hour; otherwise, he/she rides his/her motorcycle at a speed of $V = 30$ km per hour. What is the mean of the time T to get to class?

2. (a) Let X be a discrete random variable with the following PMF:

$$p_X(x) = \begin{cases} \frac{1}{9}, & if x \text{ is integer in } \{-4, 4\}, \\ 0, & \text{otherwise.} \end{cases}$$

   Consider a new random variable Z, defined by the transformation g such that $Z = g(X) = X^2$.
   1. Find the PMF of Z.
   2. Find the expectation of Z .
   3. Find the variance of Z.

   (2+1+1)

3. (a) The joint probability function of two discrete random variable X and Y is given by $P_{XY}(x, y) = c(2x + y)$, where x and y can assume all integers such that $0 \leq x \leq 2$, $0 \leq y \leq 3$, and $P_{XY}(x, y) = 0$ otherwise
   1. Find $P(X \geq 1, Y \leq 2)$
   2. Find the marginal probability function of X and Y.
   3. Find that the random variable X and Y are dependent or independent?
   4. Find $P_{XY}(y|2)$.
   5. Find $P_{XY}(Y = 1|X = 2)$.